

Network Fundamentals

What is a Network?

- ➔ Network is a collection of two or more computers or devices that are interconnected each other so that they are communicated and share the data and resources.

Types of Networks:

- ➔ Networks are mainly classified based on the geographical area that they are covered. They are different types
 1. PAN (Personal Area Network)
 2. LAN (Local Area Network)
 3. WLAN (Wire Less LAN)
 4. MAN (Metropolitan Area Network)
 5. WAN (Wide Area Network)
 6. SAN (Storage Area Network)
 7. CAN (Camps Area Networks) etc.

PAN:

- ➔ PAN stands for Personal Area Network is a network that connects the devices within small areas or the individual persons.
Ex. connecting mobile phones to ear buds, smart watches etc.

LAN:

- ➔ LAN means Local Area Network is a type of network that connects the computers and devices within the small geographical areas such as home, office, computer labs or college buildings etc.
- ➔ LANS are two types
 - ❖ Wired LANs
 - ❖ Wireless LANs

MAN:

- ➔ MAN means Metropolitan Area Network is another type of network that connects the computers and multiple LANs within a cities or large geographical areas.

WAN:

- Wide Area Network is another important network that connects the multiple MANs within a large cities or countries or around the world.

SAN:

- **SAN** means Storage Access Network which very high speed dedicated network that connects the servers to access the storage data over the internet.

CAN:

- Campus Area Network is a network that connects the different LANs within the organization, Colleges, or Companies etc.

Network Devices:

- Network devices are the hardware devices that are used to connect the computers and other resources in order to manage and control communication between them.
- Some of the network devices are Hub, Switch, Router, Modem, Repeater, Bridge, Gateway, and Access Point (AP).

Hub:

- Hub is a network device that connects the multiple devices and broadcast the received data to all the devices that are connected in the network.
- Hub does not know intended destination device in the network hence it sends the data to all the devices in the network.

Switch:

- Switch is also a network device that connects the multiple devices and sends the data to the intended destination device.
- Switch can use **MAC** address to send the data.

Router:

- Router is a network device that connects the multiple devices in the networks and it forwards data in the form of the packets between them.

Modem:

- Modem is another network device that connects the local network and internet service provider by converting the signals between network and communication medium.

Repeater:

- Repeater is another important network device that receives the weak signals and regenerate it and retransmit to extend the network range.
Ex: Wi-Fi repeater can extend the Wi-Fi signals from floor to building.

Bridge:

- Bridge is another network device that connects the two local networks and controls the traffic in between them.

Gateway:

- Gateway is a network device that acts as entry and exit point between the different networks or a systems and allowing them to communicate.

Access Point (AP):

- Access Point is a network device that allows the wireless network to connect with the wired Wi-Fi networks.

Network Topologies:

- Network topology is the physical or logical arrangement of computers, devices and other communication link,s in a network.
- Network topologies are many types they are,
 - ❖ Bus Topology
 - ❖ Star Topology
 - ❖ Ring Topology
 - ❖ Mesh Topology
 - ❖ Tree Topology
 - ❖ Hybrid Topology

Bus Topology:

- ➔ In the bus topology all the computers and devices are connected each other via single physical wire also called as main wire or backbone wire.
- ➔ But in the bus topology when main wire got damaged or failed then total network get affected.

Star Topology:

- ➔ In star topology all the computers and devices are connected to centralized device known as **switch** rather than connected with single wire
- ➔ In the star topology, when one computer fails another will be working but when central device such as switch fail the total network will affected.

Ring Topology:

- ➔ In ring topology all the devices are connected to two other devices forming a ring.
- ➔ In ring topology, data will be transmitted over ring format. So the transmission of the data will be goes to target in very slow.

Mesh Topology:

- ➔ In mesh topology, each device is to connected to all other devices in the network through the separate links .
- ➔ Data can transmit to different path to reach the destination. When one link fails other links can used to transmit the data.

Tree Topology:

- ➔ In tree topology, all the devices are connected in a hierarchical structure forming as branches.

Hybrid Topology:

- ➔ In Hybrid topology, we are using two or more topologies to form a network.